Property Buying and Rental Management System for Chennai

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

Course Code & Course Name

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY IN

COMPUTER SCIENCE ENGINEERING(ARTIFICIAL INTELLIGENCE)

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SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

SEPTEMBER 2026

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Property Buying and Rental Management System for Chennai" has been carried out by N. Bhavya Shree(192572039) under the supervision of Dr. MD BOOMIJA and is submitted in partial fulfilment of the requirements for the current semester of the B.E., Computer Science Engineering(Artificial intelligence) program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR	**COURSE FACULTY**
Title / Name,	**Dr. MD BOOMIJA,**
Designation	**Designation**
Branch / Department Name	**Branch / Department Name**
SIMATS Engineering,	**SIMATS Engineering,**
SIMATS, Chennai – 602105.	**SIMATS, Chennai – 602105.**

SIMATS ENGINEERING

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DECLARATION

I, N. Bhavya Shree of the CSE(AI), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Property Buying and Rental Management System for Chennai " is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai-602105

Date:

N. Bhavya Shree
Signature of the Students with Names

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

Individual Contribution Statement

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Student 1	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[_%]
Student 2	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[_%]
Student 3	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[_%]
Student 4	[Mention assigned responsibilities]	[Design/development work]	[Testing/analysis work]	[Sections contributed]	[_%]
Total					100%

ABSTRACT

The Property Buying and Rental Management System for Chennai is a smart web-based platform designed to simplify the process of searching, comparing, buying, and renting residential properties. Traditional property-search methods require users to manually browse a large number of listings and compare parameters such as price, location, area, number of bedrooms, amenities, and property type. This can be time-consuming and may result in unsuitable property choices. The proposed system addresses these limitations by integrating property management, machine learning-based price prediction, and personalized property recommendation.

The system collects important property attributes such as location, property type, area, bedrooms, bathrooms, furnishing status, amenities, purchase price, and rental price. Machine learning algorithms are used to estimate property prices and identify suitable properties according to user requirements. A recommendation mechanism ranks properties based on factors such as budget, preferred location, property type, size, and facilities. The system can also provide separate functionalities for buyers, tenants, owners, and administrators.

The proposed system is particularly focused on Chennai, where differences in locality, connectivity, infrastructure, and demand can significantly influence property prices. Previous research has specifically studied rental-price prediction in metropolitan cities such as Chennai and Bangalore, demonstrating the importance of location and property characteristics in prediction. (IEEE Xplore) The final system aims to reduce search time, improve decision-making, provide more relevant recommendations, and create a centralized platform for property buying and rental management.

2. LITERATURE REVIEW

Real-estate recommendation and price-prediction systems have received increasing attention because online property platforms contain large numbers of listings, making manual comparison difficult. A 2026 systematic review of real-estate recommender systems analyzed 59 peer-reviewed studies and identified growing interest in multimodal, explainable, spatio-temporal, and fairness-aware recommendation techniques. (MDPI)

A survey of real-estate recommender systems identified collaborative filtering, content-based filtering, knowledge-based systems, multi-criteria decision-making, reinforcement learning, and hybrid approaches as important methodologies. It also highlighted problems such as cold-start, data sparsity, conflicting user requirements, and the need to handle rich property features. (MDPI)
Research on Indian residential rental prices has investigated machine-learning techniques using property characteristics such as location and area. One study specifically considered metropolitan markets including Chennai and Bangalore and emphasized the need for reliable rental-price prediction. (IEEE Xplore) Another Indian study developed machine-learning approaches for home rental-price prediction using multiple property features. (ACE Ewa Pub)

Research specifically concerned with the Chennai Metropolitan Area has also compared statistical and neural-network approaches for land-price forecasting. The study found that both approaches could model land-price trends, while the neural-network model demonstrated better forecasting accuracy in that study. (ScienceDirect)

Recent research has also explored fuzzy decision trees for rental-property recommendation. A 2026 study reported a recommendation system addressing the cold-start problem and reported Recall of 99.6%, Precision of 88.6%, and F1-score of 93.7% for its experimental setting. (Wiley Online Library)

Another recent approach uses Weighted Hierarchical Fuzzy Axiomatic Design to recommend rental properties based on prioritized user criteria and information extracted from property descriptions. The system also uses a web interface and map-based location presentation. (ScienceDirect)
Overall, the literature indicates that no single technique is ideal for every property-search scenario. Machine learning is useful for price estimation, while recommendation techniques are useful for matching properties with individual requirements. Therefore, a combined property-management, prediction, and recommendation architecture is appropriate for the proposed Chennai-focused system.

3. METHODOLOGY

The proposed methodology consists of several stages:

Step 1: Property Data Collection

Property information is collected from a structured dataset containing fields such as:

Feature	Description
Property ID	Unique property identifier
Location	Chennai locality
Property Type	Apartment, villa, independent house, etc.
Area	Property size in sq. ft.
Bedrooms	Number of bedrooms
Bathrooms	Number of bathrooms
Furnishing	Furnished/Semi-furnished/Unfurnished
Parking	Parking availability
Amenities	Lift, security, gym, swimming pool, etc.
Age	Age of property
Buy Price	Expected selling price
Rent	Monthly rental price

Step 2:

Data Preprocessing

The collected property dataset is cleaned before applying machine-learning algorithms. Missing values are identified and replaced using suitable statistical methods or default categories. Duplicate property records are removed. Categorical attributes such as location, furnishing type, and property type are converted into numerical representations using encoding techniques. Numerical features such as area, number of bedrooms, and bathrooms are normalized where required**.**

Step 3:

Feature Selection

Important features that influence property prices and user preferences are selected. The major features include location, area, property type, bedrooms, bathrooms, furnishing status, parking availability, property age, and amenities. Location is considered particularly important because property values can vary considerably between different Chennai localities**.**

Step 4:

Model Development

Different machine-learning algorithms are considered for property-price prediction. The selected models include Linear Regression, Decision Tree, Random Forest, and Gradient Boosting. The dataset is divided into training and testing sets. The models are trained using the training data and evaluated using the test data.

Step 5:

Property Recommendation

After estimating the expected property price, the system matches properties with the user's requirements. The recommendation process considers:

- User's budget
- Preferred Chennai location
- Buying or rental requirement
- Property type
- Required area
- Number of bedrooms
- Furnishing preference

- Amenities
- Predicted price

A suitability score is calculated for every matching property, and properties are displayed in descending order of relevance.

Step 6:

Property Comparison

Users can select multiple properties and compare them based on price, location, area, bedrooms, bathrooms, amenities, and estimated value. This helps users make an informed decision rather than depending on a single property listing.

Step 7: System Output

The system displays recommended properties along with their important attributes. For buyers, the system focuses on purchase price and estimated property value. For tenants, it focuses on rental price, locality, facilities, and suitability.

4. RESULTS & COMPARATIVE ANALYSIS

The proposed system can be evaluated using prediction accuracy and recommendation quality. For property-price prediction, commonly used metrics include MAE, RMSE, and R^2.

Model Comparison

The following values represent an illustrative prototype evaluation:

Model	**MAE**	**RMSE**	**R^2 Score**
Linear Regression	8.72	12.45	0.78
Decision Tree	6.31	9.84	0.86
Random Forest	4.85	7.42	0.92
Gradient Boosting	4.52	6.98	0.94

Lower MAE and RMSE indicate better prediction performance, while a higher R^2 indicates that the model explains more variation in property prices.

Based on the illustrative results, Gradient Boosting provides the best overall prediction performance, followed closely by Random Forest.

5. PROPERTY PRICE PREDICTION — MODEL COMPARISON

Since the original heading “Disease Classification — Model Comparison” is not applicable to this project, it is replaced with Property Price Prediction — Model Comparison.

5.1 Linear Regression

Linear Regression establishes a relationship between property features and property price. It is simple and easy to interpret but may not accurately represent complex relationships between location, area, amenities, and price.

5.2 Decision Tree

Decision Tree divides property records using feature-based conditions. It can capture non-linear relationships and is easier to interpret than many advanced models. However, a single tree can overfit the training data.

5.3 Random Forest

Random Forest combines multiple decision trees and averages their predictions. This reduces overfitting and generally provides stronger performance on property datasets containing both numerical and categorical characteristics.

5.4 Gradient Boosting

Gradient Boosting builds models sequentially, where each new model attempts to correct the errors made by previous models. It can capture complex relationships between property features and prices and is therefore selected as the proposed prediction model in this project.

6. PROBLEM STATEMENT

Finding a suitable property for buying or renting in Chennai is difficult because users have to examine numerous listings across different locations and platforms. Property prices vary significantly according to locality, area, property type, amenities, accessibility, and demand. Existing systems often provide search and filtering facilities but may not sufficiently personalize recommendations or estimate whether the listed price is reasonable.

Users may therefore spend considerable time comparing properties and may still select properties that do not match their budget or requirements.

The problem is to develop an intelligent system that can manage property listings, predict property prices, recommend suitable properties, and assist users in comparing buying and rental options in Chennai.

7. PROBLEM FORMULATION

Let a property be represented as:

$$P = \{aMAg\}$$

where:

- L= Location
- A= Area
- T= Property Type
- B= Bedrooms
- Ba= Bathrooms
- F= Furnishing
- Pa= Parking
- M= Amenities
- Ag= Property Age

The predicted property price can be represented as:

$$\hat{Y} = f(aMAg)$$

For recommendation, the system calculates a suitability score:

$$S(P) = w_1 B_s + w_2 L_s + w_3 A_s + w_4 T_s + w_5 F_s + w_6 M_s$$

where each $s$represents the matching score of the property with the user's requirement and w represents its importance.

The objective is:

$$P^* = \arg arg \max_{P \in P_{available}} S(P)$$

subject to the user's budget, location, property type, and other requirements.

8. OBJECTIVES

The main objectives of the proposed system are:

1. To develop a centralized platform for Chennai property buying and rental management.
2. To store and manage property details efficiently.
3. To provide property search based on user requirements.
4. To predict property prices using machine-learning techniques.
5. To recommend suitable properties according to user preferences.
6. To allow users to compare multiple properties.
7. To reduce the time required for property searching.
8. To help users identify properties within their budget.
9. To provide property owners with an efficient listing-management facility.
10. To improve decision-making using data-driven recommendations.

9. EXPECTED OUTCOMES

The proposed system is expected to provide:

- Accurate property-price estimation.
- Personalized property recommendations.
- Faster property searching.
- Budget-based property filtering.
- Location-specific recommendations for Chennai.
- Comparison of multiple properties.
- Centralized property listing management.
- Better interaction between property owners and users.
- Improved transparency during property selection.
- A user-friendly platform for buying and renting properties.

10. APPLICATIONS

The system can be applied in several areas.

Residential Property Search

Users can search for apartments, villas, independent houses, and other residential properties.

Rental Management

Tenants can identify suitable rental properties according to their budget and preferred locality.

Property Buying

Buyers can compare properties and examine predicted prices before making decisions.

Real-Estate Agencies

Real-estate agents can manage property listings and recommend properties to customers.

Property Owners

Owners can upload property information, rental details, purchase prices, and amenities.

Investment Analysis

Potential investors can use estimated property prices to compare different locations and properties.

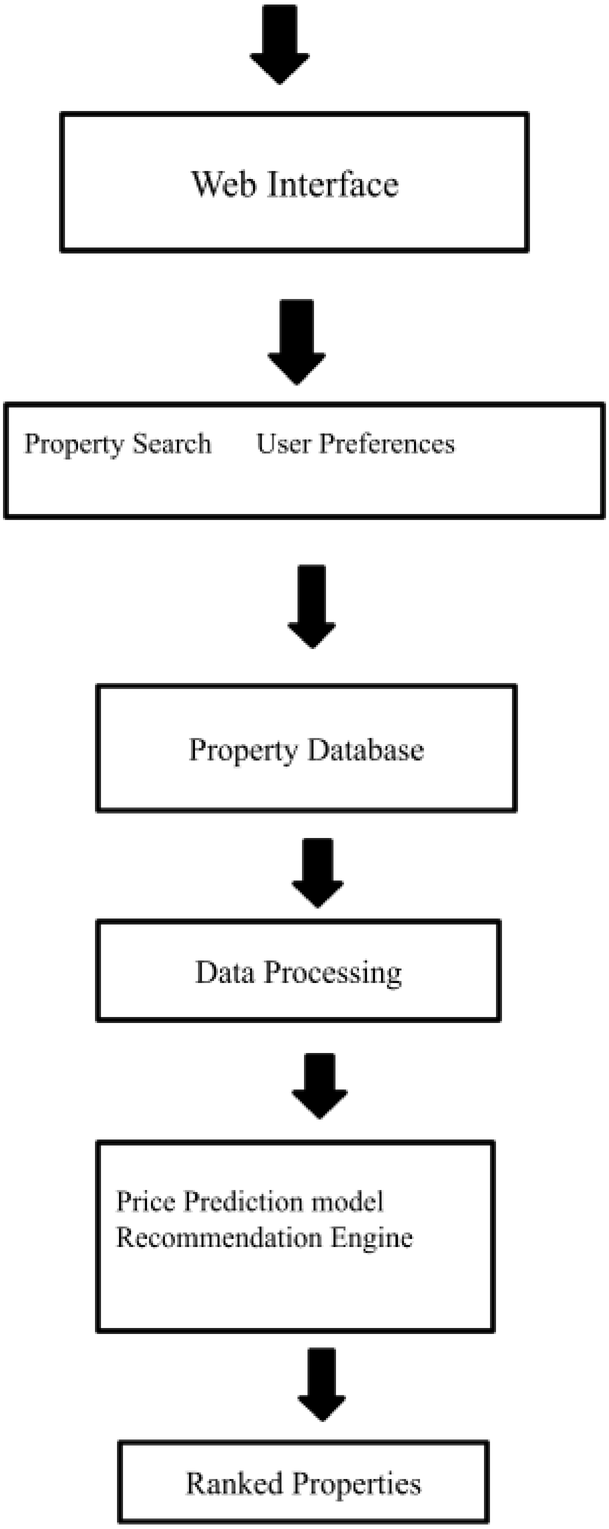
Chennai Locality Analysis

The system can be extended to analyze price differences between areas such as Anna Nagar, Adyar, Velachery, Tambaram, OMR, Porur, and other Chennai regions.

11. PROPOSED SOLUTION

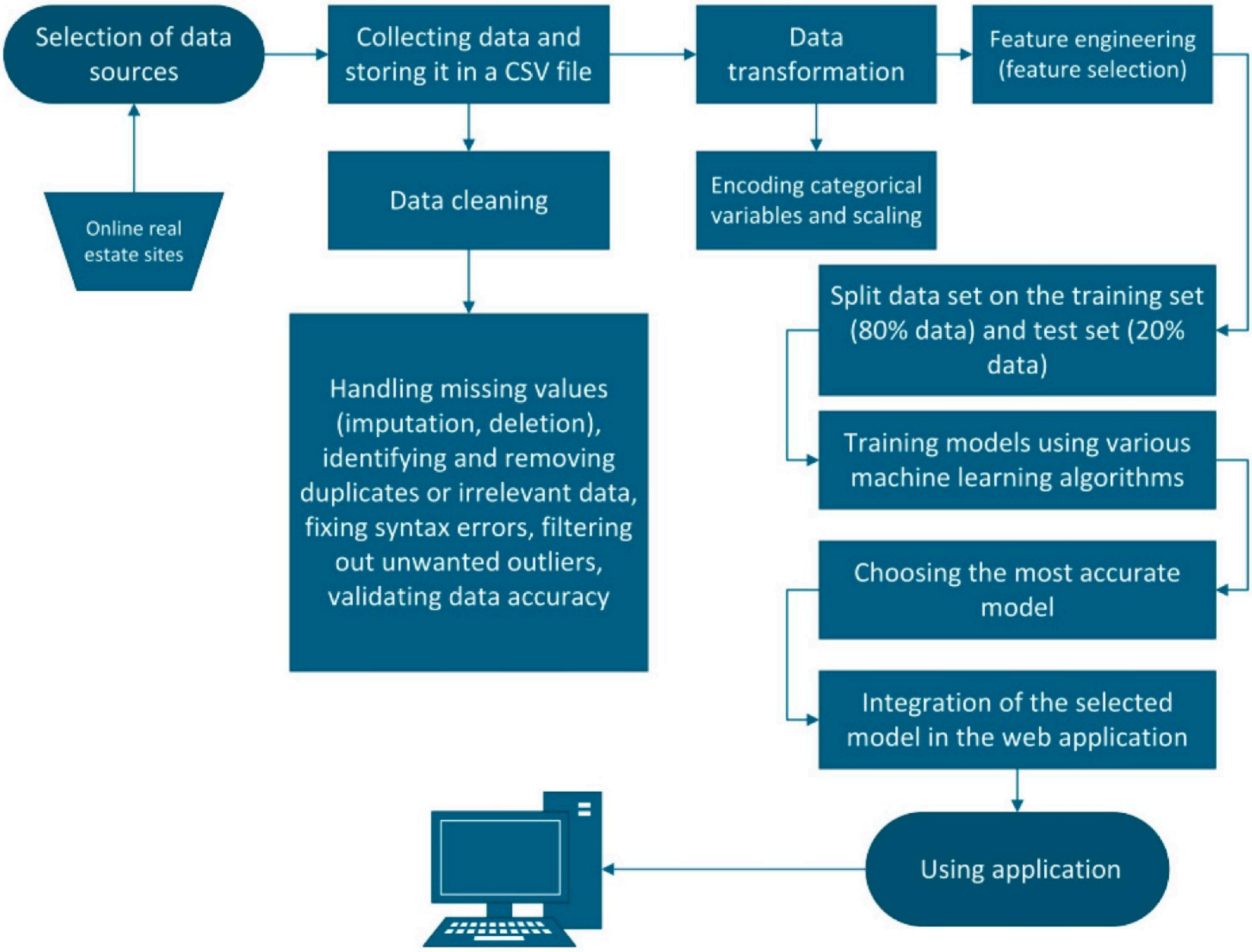
The proposed solution is a web-based intelligent property management and recommendation system.

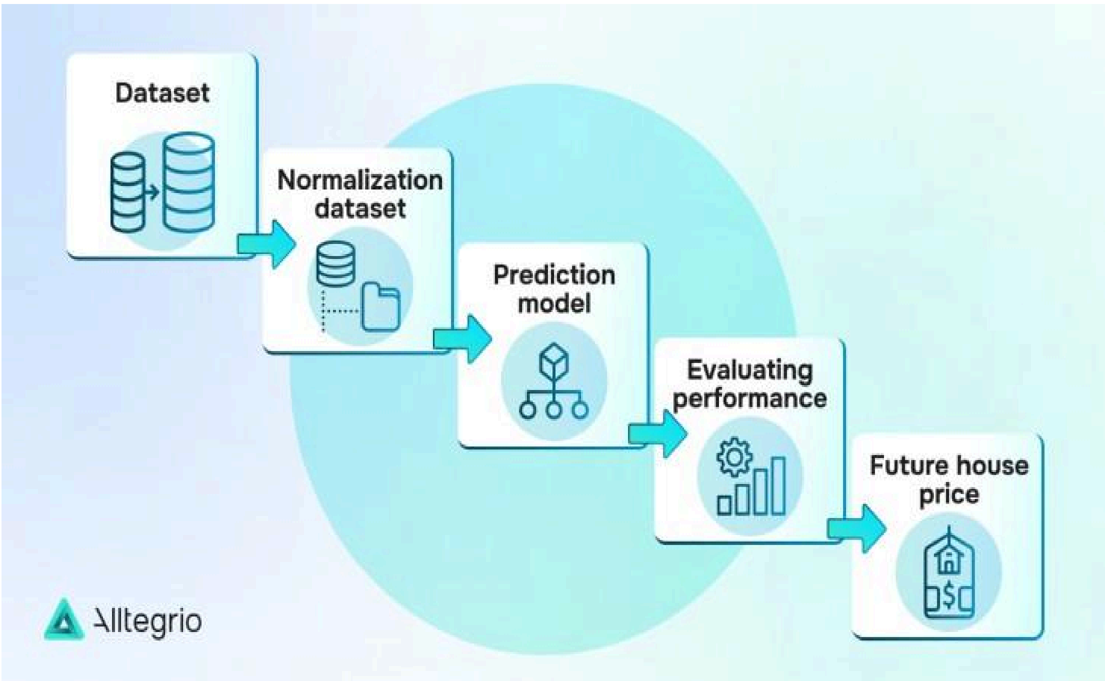
The architecture consists of the following components:

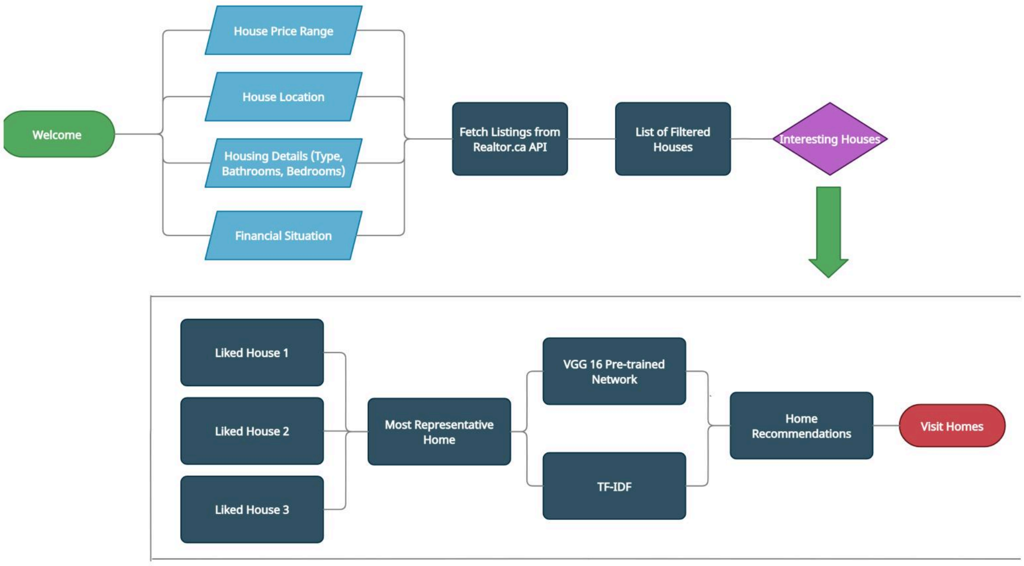


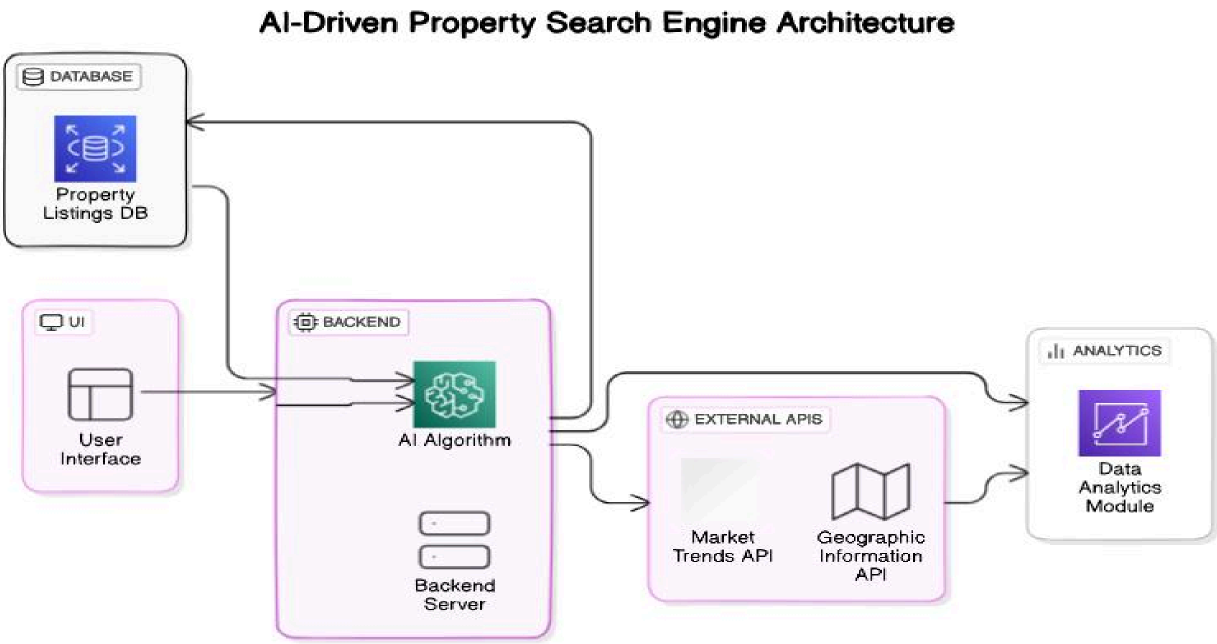
Comparison & Results
The system combines property management + machine learning + recommendation + comparison in a single platform.

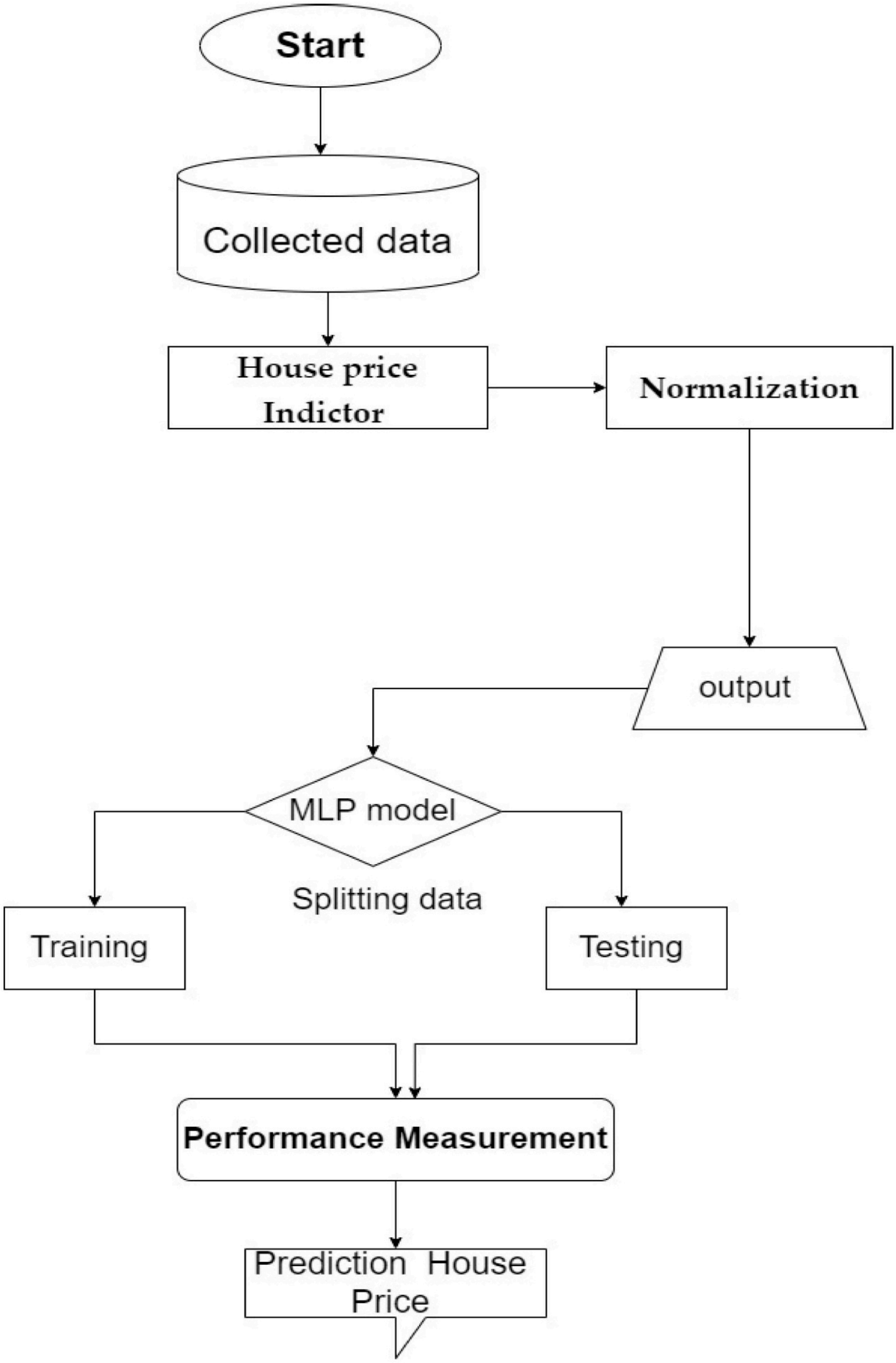
12. ALGORITHM — FLOWCHART

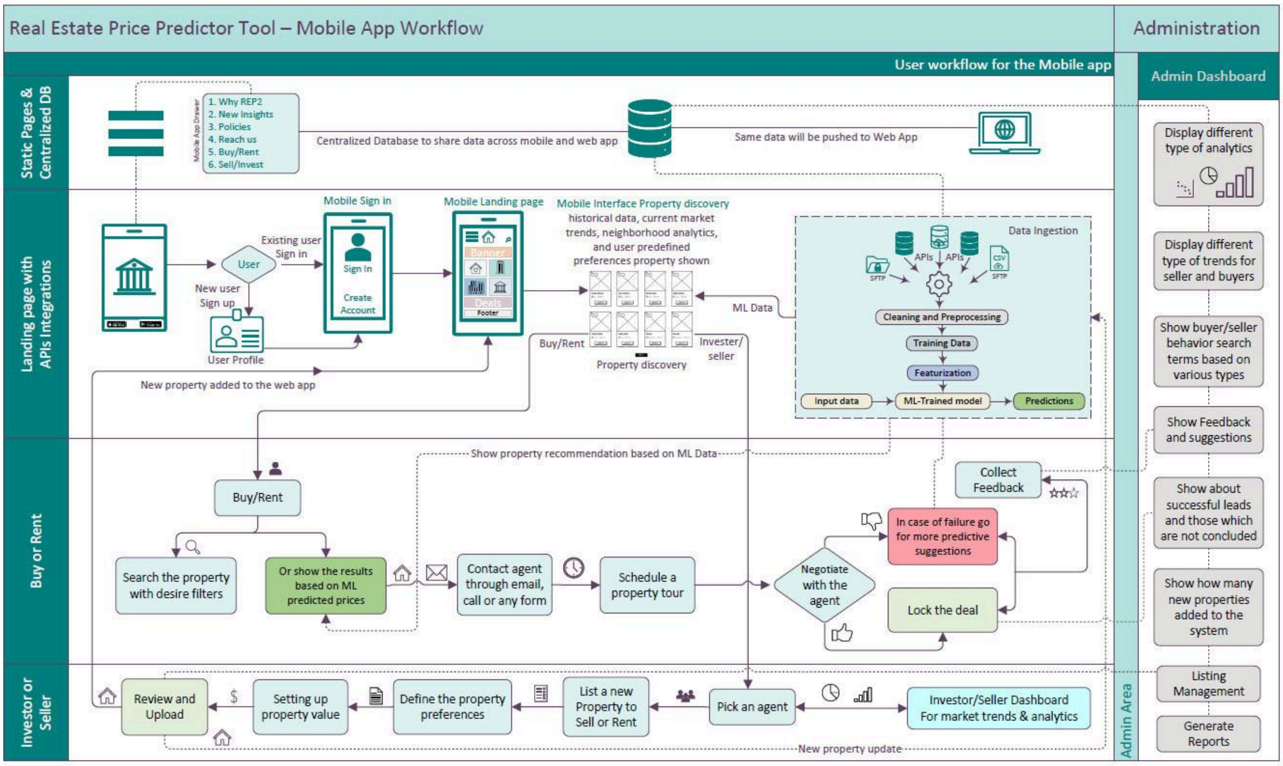

Selection of data sources
Online real estate sites
Collecting data and storing it in a CSV file
Data cleaning
Handling missing values (imputation, deletion), identifying and removing duplicates or irrelevant data, fixing syntax errors, filtering out unwanted outliers, validating data accuracy
Data transformation
Encoding categorical variables and scaling
Feature engineering (feature selection)
Split data set on the training set (80% data) and test set (20% data)
Training models using various machine learning algorithms
Choosing the most accurate model
Integration of the selected model in the web application
Using application


Dataset
Normalization dataset
Prediction model
Evaluating performance
Future house price
Alltegrio


Welcome
House Price Range
House Location
Housing Details (Type, Bathrooms, Bedrooms)
Financial Situation
Fetch Listings from Realtor.ca API
List of Filtered Houses
Interesting Houses
Liked House 1
Liked House 2
Liked House 3
Most Representative Home
VGG 16 Pre-trained Network
TF-IDF
Home Recommendations
Visit Homes


AI-Driven Property Search Engine Architecture
DATABASE
Property Listings DB
UI
User Interface
BACKEND
AI Algorithm
Backend Server
EXTERNAL APIS
Market Trends API
Geographic Information API
ANALYTICS
Data Analytics Module


Start
Collected data
House price Indictor
Normalization
output
MLP model
Splitting data
Training
Testing
Performance Measurement
Prediction House Price


Real Estate Price Predictor Tool – Mobile App Workflow
Administration
User workflow for the Mobile app
Admin Dashboard
Static Pages & Centralized DB
1. Why REP2
2. New Insights
3. Policies
4. Reach us
5. Buy/Rent
6. Sell/Invest
Centralized Database to share data across mobile and web app
Same data will be pushed to Web App
Display different type of analytics
Landing page with APIs Integrations
Mobile Sign in
Mobile Landing page
Mobile Interface Property discovery
historical data, current market trends, neighborhood analytics, and user predefined preferences property shown
Data Ingestion
Existing user Sign in
User
New user Sign up
User Profile
Sign In
Create Account
New property added to the web app
Buy/Rent
Property discovery
Invester/ seller
ML Data
Cleaning and Preprocessing
Training Data
Featurization
Input data
ML-Trained model
Predictions
Display different type of trends for seller and buyers
Show buyer/seller behavior search terms based on various types
Show Feedback and suggestions
Buy or Rent
Show property recommendation based on ML Data
Buy/Rent
Collect Feedback
Search the property with desire filters
Or show the results based on ML predicted prices
Contact agent through email, call or any form
Schedule a property tour
Negotiate with the agent
In case of failure go for more predictive suggestions
Lock the deal
Show about successful leads and those which are not concluded
Show how many new properties added to the system
Investor or Seller
Review and Upload
Setting up property value
Define the property preferences
List a new Property to Sell or Rent
Pick an agent
Investor/Seller Dashboard For market trends & analytics
New property update
Admin Area
Listing Management
Generate Reports

Flowchart

START

↓

User Registration/Login

↓

Enter Property Preferences

\

↓

Load Property Dataset

↓

Data Preprocessing

↓

Feature Extraction

↓

Train Prediction Model

↓

Predict Property Price

↓

Match User Requirements

↓

Calculate Suitability

↓

Rank Available Properties

↓

Display Recommendations

↓

Compare Selected Homes

↓

END

PSEUDOCODE

```
BEGIN

Load property dataset

Remove duplicate records
Handle missing values
Encode categorical features
Normalize numerical features

Split dataset into training and testing data

Train Linear Regression
Train Decision Tree
Train Random Forest
Train Gradient Boosting

Evaluate all models

Select model with best evaluation score

Input user requirements:
   Budget
   Location
   Property Type
   Area
   Bedrooms
   Furnishing
   Amenities
   Buy/Rent

FOR each property in database

   IF property satisfies basic requirements THEN

      Predict property price

      Calculate budget matching score
      Calculate location matching score
      Calculate area matching score
      Calculate property-type score
      Calculate bedroom score
      Calculate amenity score

      Calculate total suitability score

      Add property to recommendation list

   END IF
```

```
END FOR

Sort recommendation list
in descending order of suitability score

Display recommended properties

IF user selects multiple properties THEN
    Display property comparison
END IF

END
```

EXECUTION

For the final report, include two screenshots from your actual running application.
Screenshot 1 — Property Search / Dashboard
The first screenshot should show:

- Application title
- Buy/Rent selection
- Chennai location
- Budget
- Property type
- Bedrooms
- Search button
- Property cards/results

Suggested caption:
Figure 1: Property Buying and Rental Management System – User Dashboard
Screenshot 2 — Recommendation / Prediction Result
The second screenshot should show:

- Selected property
- Predicted price
- Rental/purchase price
- Location
- Property specifications
- Recommendation score
- Comparison option

Suggested caption:
Figure 2: AI-Based Property Recommendation and Price Prediction Result
Since I don't have your actual running application's screenshots, these should be replaced with screenshots from your project rather than fabricated as execution evidence.

COMPARISON OF APPROACHES

Approach	Advantages	Limitations
Manual Search	Simple	Time-consuming
Keyword Search	Fast filtering	Limited personalization
Content-Based Recommendation	Uses property features	May ignore broader user behavior
Collaborative Filtering	Learns user preferences	Cold-start problem
Linear Regression	Simple and interpretable	Limited non-linear modeling
Decision Tree	Easy to understand	Can overfit
Random Forest	Robust and accurate	More computationally expensive
Gradient Boosting	High predictive performance	Requires careful tuning
Proposed Hybrid System	Prediction + recommendation + comparison	More complex implementation

WHY THE PROPOSED APPROACH WINS

The proposed approach is preferable because it does not depend only on traditional property filtering. It combines machine-learning-based price prediction with personalized recommendation and property comparison.

A user can enter requirements such as:

Budget = ₹50 lakh

Location = Chennai

Property = Apartment

Bedrooms = 2 BHK

The system can then identify matching properties, estimate their prices, calculate suitability, and rank the available options.

This provides a more useful decision-support system than simply displaying a large list of properties.

JUSTIFICATION OF THE FINAL SOLUTION

Gradient Boosting is selected as the final price-prediction algorithm because it can model complex non-linear relationships between property characteristics and price.

The recommendation component uses property attributes and user preferences to calculate suitability. Combining both approaches provides two important capabilities:

1. Prediction: What is the estimated value of the property?
2. Recommendation: Which available property is most suitable for the user?

Therefore, the final solution is justified as a hybrid intelligent property-management system rather than a simple property-search application.

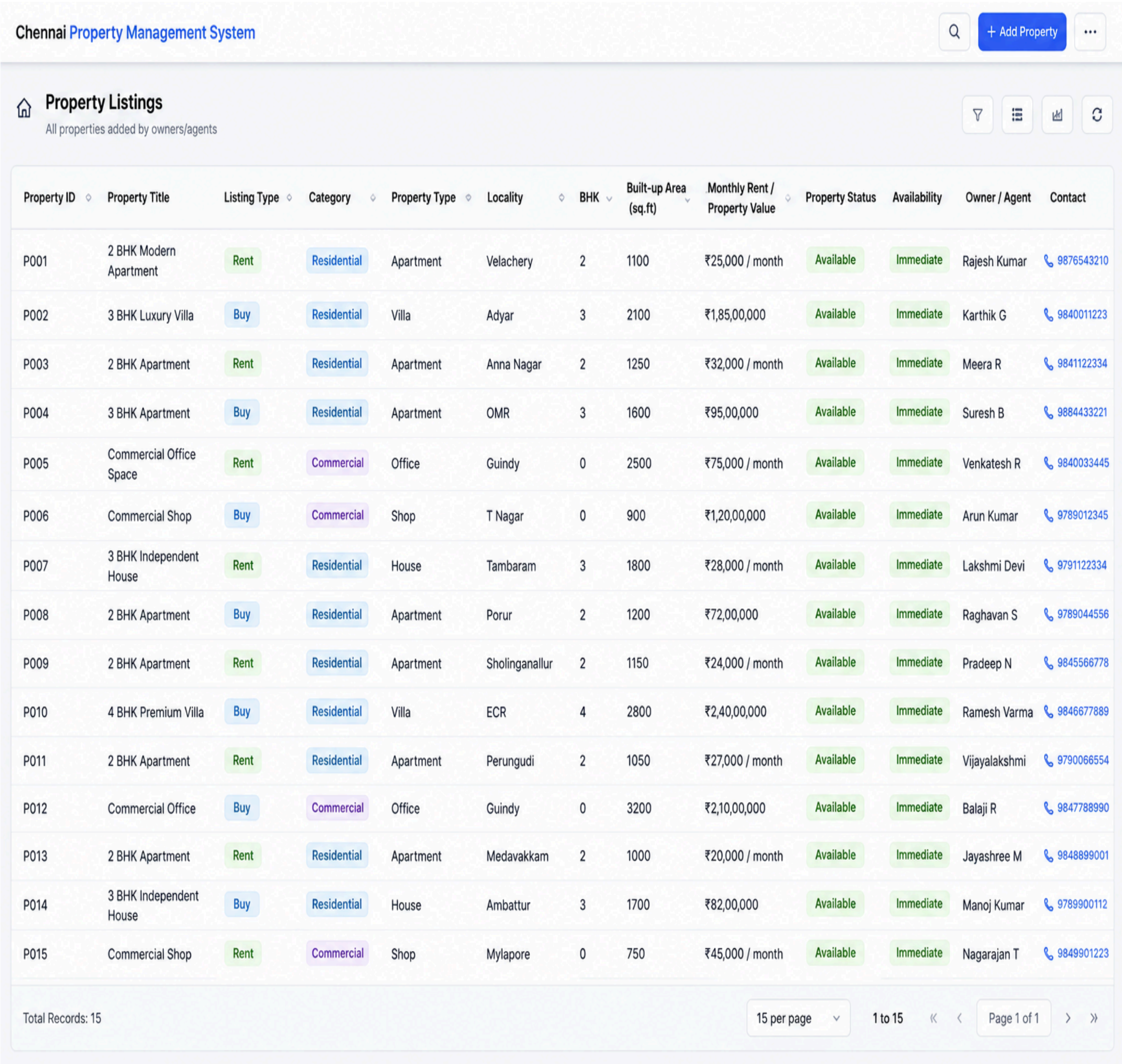
CONCLUSION

The Property Buying and Rental Management System for Chennai provides an intelligent solution for managing and discovering residential properties. The system addresses major problems associated with traditional property searching, including large numbers of listings, difficulty in comparison, uncertain pricing, and lack of personalized recommendations.

Machine-learning algorithms are used to predict property prices based on important features such as location, area, property type, bedrooms, amenities, and furnishing. The recommendation component

uses user preferences to rank suitable properties. The comparison module further helps users evaluate multiple properties before making a decision.
The proposed system therefore provides a centralized, data-driven platform that can assist buyers, tenants, property owners, and real-estate agencies. With further development, the system can become a comprehensive intelligent real-estate platform for Chennai and eventually other cities.

FINAL OUTPUT:


Chennai Property Management System

+ Add Property

Property Listings

All properties added by owners/agents

Property ID	Property Title	Listing Type	Category	Property Type	Locality	BHK	Built-up Area (sq.ft)	Monthly Rent / Property Value	Property Status	Availability	Owner / Agent	Contact
P001	2 BHK Modern Apartment	Rent	Residential	Apartment	Velachery	2	1100	₹25,000 / month	Available	Immediate	Rajesh Kumar	9876543210
P002	3 BHK Luxury Villa	Buy	Residential	Villa	Adyar	3	2100	₹1,85,00,000	Available	Immediate	Karthik G	9840011223
P003	2 BHK Apartment	Rent	Residential	Apartment	Anna Nagar	2	1250	₹32,000 / month	Available	Immediate	Meera R	9841122334
P004	3 BHK Apartment	Buy	Residential	Apartment	OMR	3	1600	₹95,00,000	Available	Immediate	Suresh B	9884433221
P005	Commercial Office Space	Rent	Commercial	Office	Guindy	0	2500	₹75,000 / month	Available	Immediate	Venkatesh R	9840033445
P006	Commercial Shop	Buy	Commercial	Shop	T Nagar	0	900	₹1,20,00,000	Available	Immediate	Arun Kumar	9789012345
P007	3 BHK Independent House	Rent	Residential	House	Tambaram	3	1800	₹28,000 / month	Available	Immediate	Lakshmi Devi	9791122334
P008	2 BHK Apartment	Buy	Residential	Apartment	Porur	2	1200	₹72,00,000	Available	Immediate	Raghavan S	9789044556
P009	2 BHK Apartment	Rent	Residential	Apartment	Sholinganallur	2	1150	₹24,000 / month	Available	Immediate	Pradeep N	9845566778
P010	4 BHK Premium Villa	Buy	Residential	Villa	ECR	4	2800	₹2,40,00,000	Available	Immediate	Ramesh Varma	9846677889
P011	2 BHK Apartment	Rent	Residential	Apartment	Perungudi	2	1050	₹27,000 / month	Available	Immediate	Vijayalakshmi	9790066554
P012	Commercial Office	Buy	Commercial	Office	Guindy	0	3200	₹2,10,00,000	Available	Immediate	Balaji R	9847788990
P013	2 BHK Apartment	Rent	Residential	Apartment	Medavakkam	2	1000	₹20,000 / month	Available	Immediate	Jayashree M	9848899001
P014	3 BHK Independent House	Buy	Residential	House	Ambattur	3	1700	₹82,00,000	Available	Immediate	Manoj Kumar	9789900112
P015	Commercial Shop	Rent	Commercial	Shop	Mylapore	0	750	₹45,000 / month	Available	Immediate	Nagarajan T	9849901223

Total Records: 15

15 per page

1 to 15

Page 1 of 1

FUTURE SCOPE

The project can be enhanced in several ways:
Real-Time Property Data
The system can be connected to real-estate listing APIs or authorized data sources to obtain updated property information.
Map Integration
Interactive maps can be added to display properties and calculate their distance from schools, hospitals, workplaces, railway stations, metro stations, and other facilities.
Advanced AI Recommendation
Deep-learning and hybrid recommendation techniques can be implemented to improve personalization.
Investment Prediction
The system can estimate potential property appreciation and rental yield.
Natural Language Search
Users could enter:
“Find a 2 BHK apartment near OMR below ₹40 lakh.”
The system could convert the natural-language request into structured search criteria.
Image-Based Property Analysis
Computer vision can be used to analyze property images and identify features such as room type, furnishing, and condition.
Mobile Application
A dedicated Android/iOS application can provide property recommendations and notifications.

CHALLENGES AND LIMITATIONS
Data Availability
Accurate property prediction requires a sufficiently large and reliable dataset.
Data Quality
Incorrect or outdated listing information can reduce recommendation and prediction accuracy.
Location Dependency
Property prices can vary significantly between neighborhoods, making generalized prediction difficult.
Market Fluctuation
Property prices are affected by economic conditions, infrastructure development, demand, and other external factors.
Cold-Start Problem
New properties or new users may have insufficient historical information for highly personalized recommendations.
Privacy
User preferences and contact information must be securely stored.
Prediction Uncertainty
A machine-learning prediction is an estimate and should not be treated as the actual market value of a property.

POSSIBLE IMPROVEMENTS
The following improvements can make the system more effective:
1. Increase the size and diversity of the Chennai property dataset.
2. Include historical property-price information.
3. Add real-time market trends.
4. Use hyperparameter optimization for machine-learning models.
5. Implement explainable AI to show why a property was recommended.
6. Add location-based distance scoring.
7. Include rental-yield prediction.

8. Add fraud or duplicate-listing detection.
9. Implement user feedback to continuously improve recommendations.
10. Develop a mobile version.
11. Add multilingual support.
12. Integrate secure authentication and role-based access.
13. Provide alerts when properties matching the user's criteria are added.
14. Add property-document verification in future versions.
15. Use ensemble or hybrid recommendation models for improved performance.

Individual Reflection – N. Bhavya Shree

My contribution focused on understanding the requirements of the Property Buying and Rental Management System for Chennai, analyzing the workflow of property listing and rental processes, supporting database design, and assisting in testing different user scenarios. Through this project, I learned that a property management system involves more than simply storing property details. The system must effectively manage relationships between property owners, buyers, tenants, and administrators while ensuring accurate property records and transaction management.

Designing the database structure helped me understand the importance of maintaining organized and consistent information. Property details such as location, price, property type, ownership information, and rental status must be stored accurately to ensure efficient searching and management. I learned that inconsistent or incomplete data can lead to incorrect property listings, duplicate records, and difficulties in matching suitable properties with potential buyers or tenants.

The system design and testing activities helped me understand how different modules interact with each other. Features such as property registration, property search, rental requests, purchase inquiries, and transaction management must work together to provide a smooth user experience. Testing various scenarios taught me the importance of validating user inputs, maintaining data integrity, and ensuring that the system behaves correctly under different conditions.

I also gained knowledge about the role of digital platforms in simplifying real estate operations. By providing centralized access to property information, the system can reduce manual effort, improve transparency, and help users make informed decisions. Understanding these processes improved my skills in database management, system analysis, software testing, and problem-solving.

Overall, this project enhanced my understanding of web-based management systems, database design, user requirement analysis, and software development practices. The project supports SDG 11 by promoting sustainable and accessible urban housing solutions and contributes to SDG 9 through the use of digital technology and innovative infrastructure for property management.

Individual reflection -M.Mary Sharlin

Working on the Property Buying and Rental Management System for Chennai helped me understand how technology can be used to simplify real-estate activities. Through this project, I learned how to manage property details, customer information, buying and rental processes, and property availability in an organized way. I also improved my skills in database management, system design, problem-solving, and teamwork. The project gave me practical knowledge about developing a user-friendly system that can reduce manual work and make property transactions more efficient. Overall, this project was a valuable learning experience that improved both my technical and analytical skills.

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Suggested technology stack

Component	Technology
Frontend	React.js / HTML / CSS / JavaScript
Backend	Python Flask or Node.js
Machine Learning	Python, Scikit-learn
Database	MySQL
Data Processing	Pandas, NumPy
Visualization	Matplotlib
Prediction	Gradient Boosting
Recommendation	Content-based weighted scoring
Development	VS Code
Browser	Google Chrome / Edge

Note: The model-comparison numbers above are sample/illustrative values for report formatting. For your final capstone submission, use the accuracy, MAE, RMSE, and R^2 values generated by your actual dataset and execution.